

Name :Roshan Saral Kumar(CLOUD-COMPUTING-INTERN)
DATE:21-10-2025

TASK 2

- **Task 2: Launch a VirtualMachine(VM) EC2 INSTANCE**
- **Objective:- To understand *how cloud virtual machines work by creating, configuring, and connecting to a cloud-based server instance using a free-tier account.**
- **This task introduces interns to *Infrastructure-as-a-Service (IaaS)* — the foundation of all cloud operations.**
- **TOOLS USED:- AWS EC2 (Free Tier)**
- **THIS IS THE SIGN IN CONSOLE PAGE AND IAM USING SIGN USING ROOT USER EMAIL (ROOT USER MEANS I HAVE ACCESS TO ALL THE SERVICES OF AWS IF IT IS AN IAM (IDENTITY AND ACCESS MANAGEMENT USER THEN THE USER WILL HAVE ACCESS TO ONLY A LIMITED NUMBER OF SERVICES THAT AWS OFFERS AND THE IAM USERS WILL HAVE IAM POLICIES ATTACHED TO THEM SO ONLY THOSE SERVICES THEY WILL BE ABLE TO USE) CURRENTLY LOGGING IN AS ROOT USER EMAIL.**

- AFTER CLICKING ON SIGN IN USING ROOT USER EMAIL I WILL GET THIS PAGE WHERE I HAVE TO ENTER MY EMAIL-ID. AND AS WE CAN SEE ROOT USER IS SELECTED BY DEFAULT. WE CAN ALSO SELECT IAM USER IN THIS PAGE IN CASE IF THERE IS A CHANGE.

Sign In

Access your AWS account by user type.

User type (not sure?)

☒ **Root user**
Account owner that performs tasks requiring unrestricted access.

☐ **IAM user**
User within an account that performs daily tasks.

Email address

Next

OR

New to AWS? Sign up

By continuing, you agree to [AWS Customer Agreement](#) or other agreement for AWS services, and the [Privacy Notice](#). This site uses essential cookies. See our [Cookie Notice](#) for more information.

Amazon Lightsail

Lightsail is the easiest way to get started on AWS

[Learn more »](#)

- AFTER ENTERING OUR EMAIL WE NEED TO GIVE OUR PASSWORD

Root user sign in ⓘ

Enter the password for
 (not you?)

Password

☐ Show password [Forgot password?](#)

Sign in

Sign in to a different account

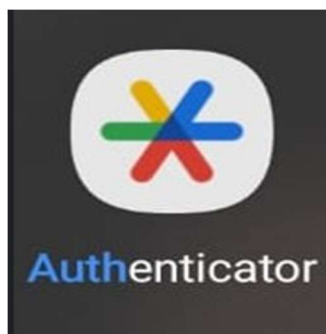
[Create a new AWS account](#)

Amazon Lightsail

Lightsail is the easiest way to get started on AWS

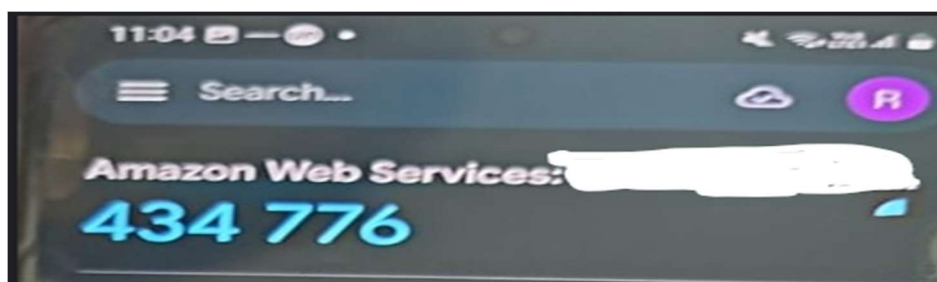
[Learn more »](#)

- **SINCE I HAVE AN AWS ACCOUNT ALREADY CREATED I ACTUALLY MADE AN MFA FOR MY ROOT USER . MFA STANDS FOR (MULTI-FACTOR AUTHENTICATION CODE) IT ACTS AS A DOUBLE LAYER PROTECTION FOR OUR ROOT USER OTHERWISE WE CAN GET HACKED SO ALREADY PROVIDE AN EXTRA LAYER OF PROTECTION TO BE ON THE SAFER SIDE.**
- **MFA CODE IS GIVEN BY TOTEM TOKENS OR U CAN ALSO USE THE “GOOGLE AUTHENTICATOR APP” IN YOUR MOBILE WHICH IS MOSTLY PREFERRED.**

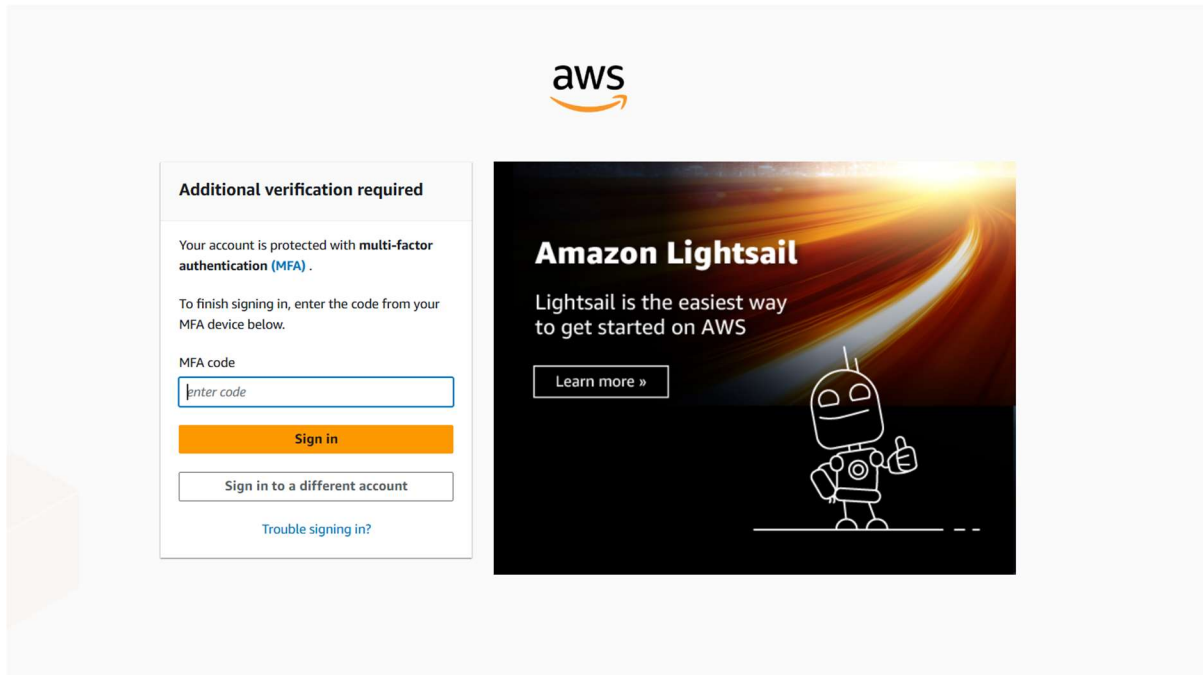


- **THIS APP IS IN MY MOBILE AND I HAVE TO GIVE THE MFA CODE WHICH IS ASKED FOR THE VERIFICATION OF ROOT USER.**
- **ONCE I HAVE CLICKED ON THIS APP I WILL SEE THE BELOW SCREEN IN MY MOBILE.**

- **THE SCREEN IN MY MOBLIE LOOKS LIKE THIS:-THE CONTENT WHICH IAM HIDING USING A WHITE COLOR IS MY PRIVATE INFORMATION OF MY ACCOUNT WHICH CANNOT BE SHARED BUT HIS IS HOW A SIMPLE MFA CODE LOOKS LIKE AND THE MFA CODE KEEPS ON CHANGING ON A DAILY BASIS.ALWAYS USE MFA TO ENSURE ENHANCED SECURITY.**



- **THIS IS THE CODE THAT I HAVE TO ENTER IN MY AWS WEB BROWSER IN THE BELOW IMAGE IT IS SHOWN.**



- **AFTER SIGNING IN WE ENTER THE AWS MANAGEMENT CONSOLE AND THE DEFAULT REGION IS NORTH VIRGINIA THAT IS “us-east-1” WE CAN SELECT DIFFERENT REIGIONS. IN THE IMAGE IT IS CLEARLY SHOWN IN THE RIGHT HAND SIDE WHICH REGION IAM CURRENTLY WORKING IN. THE BELOW IMAGE SHOWS THE SERVICES WHICH I VISITED AND THE CURRENT REGION WHICH IAM WORKING IN.**
- **TERMINOLOGIES:-**
- **Region:-** It is a geothermal location which consists of a group of AZs (AVAILABILITY ZONES) and the network of a region is called a VPC(VIRTUAL PRIVATE CLOUD).
- **AZ(Availability Zone):-** they are a group of data centers.the network of a Az is a subnet.
- **Edge Location:-** it is a location in between various Azs or it is found at the boundary of an az In order to cache the incoming data acts like a storage for easier retrieval.

PLS NOTE CHANGE STARTS FROM HERE BELOW:-

- **NOTE:-WE ARE WORKING IN THE PUBLIC SUBNET THAT IS “ap-south-1”** if we are working in any azs it will show ap-south-1a,ap-south-1b and so on. We can check the azs in our aws management console itself that is given in below figure.

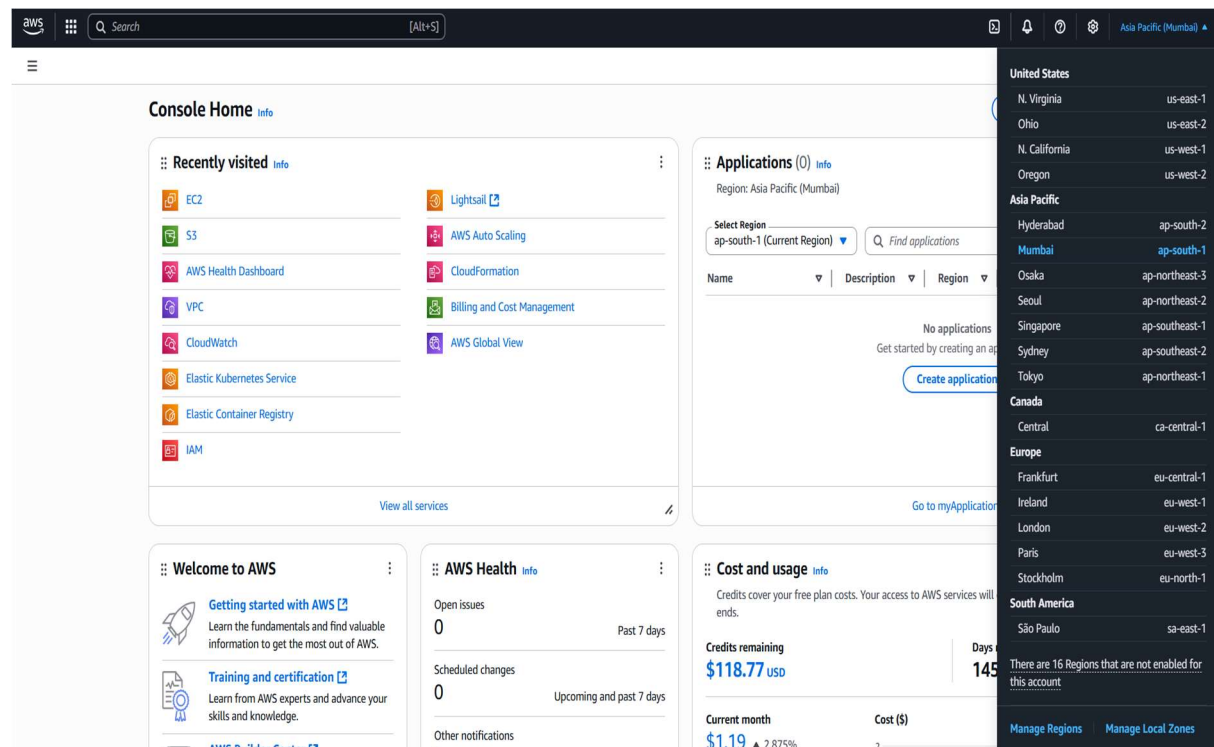
HERE WE WILL BE ABLE TO SEE THE SUBNETS IN THE VPC DASHBOARD IF WE CLICK ON SUBNETS THEN TOTALLY THERE ARE 3 SUBNETS THAT IS PRIVATE

SUBNETS FROM “ap-south-1a” to” ap-south-1c” totally we have 3 private subnets or we can also say we have 3 azs under Asia pacific (MUMBAI) REGION THAT IS “AP-SOUTH-1” REGION. This is provided by aws itself by default we cannot create any Availability Zones.

☐	Canada (Calgary)	⊗ Disabled
☐	Europe (Zurich)	⊗ Disabled
☐	Europe (Milan)	⊗ Disabled
☐	Europe (Spain)	⊗ Disabled
☐	Israel (Tel Aviv)	⊗ Disabled
☐	Middle East (UAE)	⊗ Disabled
☐	Middle East (Bahrain)	⊗ Disabled
☐	Mexico (Central)	⊗ Disabled
☐	Asia Pacific (Hyderabad)	ⓘ Enabling
☒	Asia Pacific (Tokyo)	✔ Enabled by default
☒	Asia Pacific (Seoul)	✔ Enabled by default
☒	Asia Pacific (Osaka)	✔ Enabled by default
☒	Asia Pacific (Mumbai)	✔ Enabled by default
☒	Asia Pacific (Singapore)	✔ Enabled by default
☒	Asia Pacific (Sydney)	✔ Enabled by default
☒	Canada (Central)	✔ Enabled by default
☒	Europe (Frankfurt)	✔ Enabled by default
☒	Europe (Stockholm)	✔ Enabled by default
☒	Europe (Ireland)	✔ Enabled by default
☒	Europe (London)	✔ Enabled by default

- **SEE IN THE ABOVE IMAGE WHAT IAM DOING IS THAT IAM ENABLING CERTAIN REGIONS WHICH ARE NOT THERE IN MY REGIONS LIST LIKE FOR EXAMPLE HYDERABAD.WHICH IS “ap-south-2”.It will take some time for aws to enable the regions based on our configurations and accounts.**

SO IF WE SEE IN THE BELOW FIGURE INITIALLY YHYDERABAD WAS NOT THERE BUT ONCE IT WAS ENABLED IT IS REFLECTING IN MY REGIONS LIST.JUST ABOVE MUMBAI.



I AM GOING TO CURRENTLY LAUNCH MY EC2 INSTANCE IN MUMBAI EVEN WE CAN LAUNCH OUR INSTANCE IN HYDERABAD ALSO BUT IAM CHOOSING MUMBAI BECAUSE OF THESE DIFFERENCES.

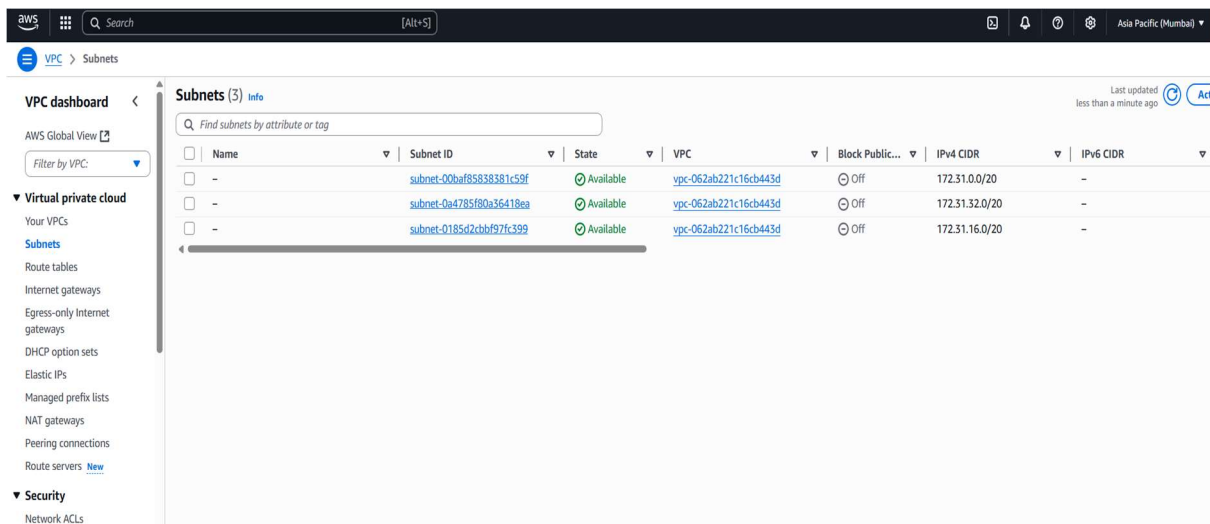
REASONS WHY I CHOSE MUMBAI (AP-SOUTH-1):-

Feature	ap-south-1 (Mumbai)	ap-south-2 (Hyderabad)
Launch Year	2016	2022
Service Coverage	✅ Full (EC2, S3, RDS, Lambda)	🟡 Growing (some services limited)
Availability Zones	3	3
Free Tier Support	✅ Yes	✅ Yes
Latency from Bangalore	✅ Lower (~20–30 ms)	🟡 Slightly higher (~30–40 ms)
Marketplace AMIs	✅ More available	🟡 Fewer options

Feature	ap-south-1 (Mumbai)	ap-south-2 (Hyderabad)
Documentation & Tutorials	✅ Widely supported	🟡 Less coverage
Enterprise Adoption	✅ Mature	🟡 Emerging

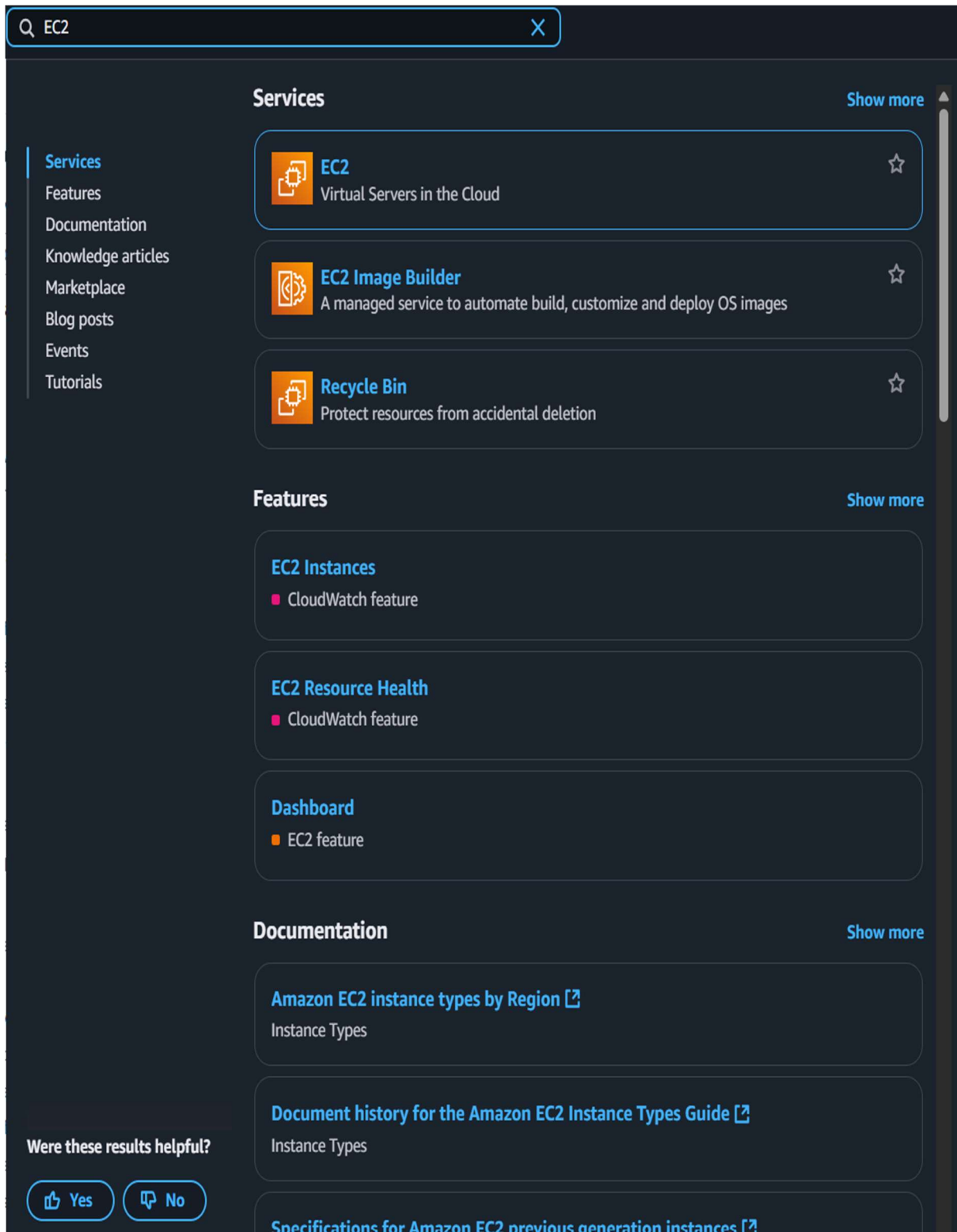
THESE ARE THE REASONS WHY I CHOSE MUMBAI.

SO CURRENTLY IAM WORKING IN AP-SOUTH-1 REGION IT HAS 3 PRIVATE SUBNETS AS SHOWN IN THE BELOW FIGURE THAT IS 3 AVAILABILITY ZONES.CLEARLY WE HAVE 3 PRIVATE SUBNETS FOR ASIA PACIFIC(MUMBAI) REGION WHICH IS THE NEAREST REGION TO BANGALORE.



- **NOW IAM GOING TO START LAUNCHING MY EC2 INSTANCE.....**

- **STEP 1:- CREATING EC2 INSTANCE:- SEARCH FOR EC2 IN THE AWS MANAGEMENT CONSOLE.**



- **STEP 2 : CLICK ON EC2 AND WE WILL GET THE BELOW PAGE:-**

- **IN THE ABOVE FIGURE “CLICK ON LAUNCH EC2 INSTANCE.”**

- **DEPENDING ON THE REQUIREMENTS GIVEN BY ELEVATE-LABS I WILL BE LAUNCHING MY INSTANCE.**

THE DETAILS ARE:-

- 1) **Choose OS: Ubuntu 22.04 LTS (lightweight and stable).(THIS IN AWS IS NOTHING BUT MY AMI IMAGE THAT IS AMAZON MACHINE IMAGE):-**

TERMINOLOGY:-

- **AMI:-**
- An **Amazon Machine Image (AMI)** is a blueprint for launching EC2 instances. It includes:
 - **Operating system** (e.g., Ubuntu, Amazon Linux)
 - **Pre-installed software** (e.g., web servers, databases)
 - **Launch permissions** (who can use it)
 - **Block device mappings** (attached storage volumes)
- You must specify an AMI when launching an EC2 instance. AWS offers:
 - **AWS-managed AMIs** (official, secure)
 - **Marketplace AMIs** (third-party, paid/free)
 - **Custom AMIs** (your own saved configurations)

DIFFERENCE BETWEEN THE AMIS :-

AMI Type	Description	Pros	Cons	Example
AWS-managed	Official AMIs provided by AWS (e.g., Ubuntu, Amazon Linux)	✓ Secure and regularly updated ✓ Free-tier eligible ✓ Widely supported	✗ Limited customization ✗ No pre-installed third-party tools	ami-0a1234567890abcdef (Ubuntu 22.04 LTS)
Marketplace	AMIs published by third-party vendors (e.g., Bitnami, Fortinet)	✓ Pre-configured with software ✓ Saves setup time ✓ Commercial support	✗ May incur extra cost ✗ Vendor lock-in ✗ Varies in quality	Bitnami WordPress AMI, Fortinet Firewall AMI
Custom	AMIs you create from your own configured EC2 instance	✓ Fully tailored to your needs ✓ Ideal for scaling identical setups	✗ Requires maintenance ✗ Storage costs ✗ Region-specific	ami-0c9b1234567890xyz (Your saved EC2 image)

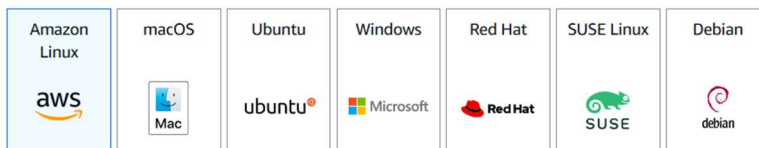
- **CURRENTLY WE ARE USING UBUNTU:22.04 LTS AMI IMAGE WHICH IS AN AWS MANAGED IMAGE FOR FREE-TIER.**
- **IF WE WANT MARKETPLACE AMIS OR COMMUNITY BASED AMIS WE CAN SEARCH NEXT TO THE AMI LIST WHICH AWS IS PROVIDING ..**

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Quick Start



[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

- **WHEN WE CLICK ON BROWSE MORE AMIS WE GET A PAGE LIKE THIS WHICH IS THE MARKET PLACE AMIS AND COMMUNITY BASED AMIS WHERE WE NEED TO PAY FOR THE USAGE OR THE REQUIREMENTS.**

Selected AMI: (ami-06fa3f12191aa3337) (Quick Start AMIs)

Search for an AMI by entering a search term e.g. "Windows"

Quick Start AMIs (45) Commonly used AMIs | My AMIs (0) Created by me | **AWS Marketplace AMIs (6082)** AWS & trusted third-party AMIs | Community AMIs (500) Published by anyone

▼ Refine results

Categories
Infrastructure (4568)
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Business Applications (848)
Industries (322)
Cloud Operations (322)
Machine Learning (304)
IoT (147)
AI Agents & Tools (93)
Professional Services (44)
Data Products (7)

▼ Publisher
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☐ Supported Images (134)
☐ cloudimg (128)
☐ Amazon Web Services (124)
☐ Hardened Images (79)
☐ Code Creator (77)
☐ Blume (77)
☐ Arara Solutions (76)
☐ Citadel Security (75)
[Show All](#)

▼ Pricing model

All products (6082 results) showing 1 - 50

Sort By: Relevance



Wowza Streaming Engine (Linux PAID)

By Wowza Media Systems, Inc. | Ver 4.9.6
★★★★☆ 6 AWS reviews | 105 external reviews

\$85.00/mo + \$0.69 to \$9.08/hr for software + AWS usage fees

Wowza Streaming Engine(TM) server software is a powerful, flexible, solution that powers reliable streaming of high-quality video and audio to any device, anywhere. Built for builders integrating video into their business, Wowza Streaming Engine software running on Amazon EC2 enables you to take...

Select



OpenVPN Access Server / Self-Hosted VPN (BYOL)

By OpenVPN Inc. | Ver 2.13.1
★★★★☆ 54 AWS reviews | 304 external reviews

If you want to purchase licenses directly from AWS, we recommend our PAYG listing instead: <https://aws.amazon.com/marketplace/pp/prodview-f5qcwyw2gy256> Access Server for AWS delivers the best-of-breed VPN solution for secure remote access, site-to-site VPN and secure SaaS access for organizations o...

Select



VM-Series Virtual Next-Generation Firewall (BYOL)

By Palo Alto Networks | Ver PAN-OS 10.2.13-H16
★★★★☆ 11 AWS reviews | 117 external reviews

VM-Series, when combined with native AWS services, enable you to create "touchless" deployments and allow your developers to operate at the speed of the cloud. VM-Series protects your applications and data using whitelisting and segmentation policies that are dynamically updated based on AWS tags...

Select

- **THE ABOVE FIGURE SHOWS MARKET PLACE AMIS AND HAS COST THEY ARE NOT FREE TIER.AND WE HAVE MANY FILTERS BASED ON USAGE AND SCENARIO WE CAN FILTER THE AMIS DEPENDING ON OUR REQUIREMENTS.**

IN THE BELOW FIGURE I HAVE CHOSEN UBUNTU AS SPECIFIED BY ELEVATE-LABS AND CREATING MY OWN INSTANCE KNOWN AS “ELEVATE_INSTANCE”. IN THE AMI IMAGE THAT IS IN THE APPLICATION AND OS IMAGES I HAVE CHOSEN UBUNTU:22.04 LTS AS MY AMI IMAGE. SHOWN IN BELOW IMAGE.

Name and tags

Name

ELEVATE_INSTANCE

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

ami-02d26659fd82cf299 (64-bit (x86)) / ami-0b9093ea0a0fed92 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture

AMI ID

Publish Date

Username

64-bit (x86)

ami-02d26659fd82cf299

2025-08-21

ubuntu

Verified provider

Summary

Number of instances

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)

ami-02d26659fd82cf299

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

IF WE CLICK ON THE DROP DOWN MENU OF UBUNTU 22.04 LTS WE HAVE MANY FREE TIER OPTIONS AND DIFFERENT VOLUME TYPES BUT CURRENTLY IAM CHOOSING THE DEFAULT CONFIGURATION DEPENDING ON THE REQUIREMENTS IT CAN CHANGE. IT IS AN SSD VOLUME TYPE IAM USING

ELEVATE_INSTANCE

Add additional tags

Q |

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

ami-02d26659fd82cf299 (64-bit (x86)) / ami-0b9093ea0a0fed92 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Ubuntu Server 22.04 LTS (HVM), SSD Volume Type

ami-07f07a6e1060cd2a8 (64-bit (x86)) / ami-0e639fddd264584f2 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Ubuntu Pro - Ubuntu Server Pro 24.04 LTS (HVM), SSD Volume Type

ami-0338ab7bf038aed53 (64-bit (x86)) / ami-0e89db42cfdca3d1d (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Ubuntu Server 22.04 LTS (HVM) with SQL Server 2022 Standard

ami-0fc92b65f5dbf615e (64-bit (x86))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Deep Learning Base AMI with Single CUDA (Ubuntu 22.04)

ami-040d9fa154ea0e208 (64-bit (x86)) / ami-016bb4cf07db96e47 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Deep Learning OSS Nvidia Driver AMI GPU PyTorch 2.8 (Ubuntu 24.04)

ami-0ef7dfd09f5e0ea3 (64-bit (x86)) / ami-02d6faa556235af4d (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Deep Learning OSS Nvidia Driver AMI GPU TensorFlow 2.18 (Ubuntu 22.04)

ami-02d26659fd82cf299 (64-bit (x86)) / ami-0b9093ea0a0fed92 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

ami-02d26659fd82cf299 (64-bit (x86)) / ami-0b9093ea0a0fed92 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

NOW HERE THERE IS ONPE PART THAT WE NEED TO NOTICE ONE THING IS THAT IF WE CAN OBSERVE CAREFULLY IN THE ABOVE IMAGE THERE ARE TWO UBUNTU SERVER 22.04 LTS AND EVERYTHING LOKKS SAME BUT THEY HAVE

SUBTLE DIFFERENCES THERE AMI IDS ARE DIFFERENT AND THE REASON BEHIND HAVING MULTIPLE AMIS ARE:-

- **Different Build Dates:** Canonical (Ubuntu's publisher) frequently updates AMIs with the latest security patches and kernel versions. These could be newer builds of the same base image.
- **Different Regions or Zones:** Sometimes AMIs are replicated across availability zones or updated for performance tuning in specific regions.
- **Marketplace vs. Community AMIs:** Even official AMIs can be published under slightly different contexts—like for EC2 Image Builder or automation pipelines.
- **Minor Version Differences:** The base OS might be 22.04 LTS, but one AMI could include a newer kernel or updated packages.

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-02d26659fd82cf299 (64-bit (x86)) / ami-0b9093ea00a0fed92 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Ubuntu Server 22.04 LTS (HVM), SSD Volume Type
ami-07f07a6e1060cd2a8 (64-bit (x86)) / ami-0e639fddd264584f2 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

NOW IN ARCHITECTURE TYPES WE HAVE TWO THAT IS:-

Architecture

64-bit (x86) ▲

64-bit (x86) ✓

64-bit (Arm)

AMI ID

ami-02d26659fd82cf299

Publish Date

2025-08-21

Username ⓘ

ubuntu

Verified provider

64 BIT(x86) and 64 BIT(ARM) The differences are:-

Feature	64-bit x86 (x86_64)	64-bit ARM (ARM64 / aarch64)
Architecture Type	CISC (Complex Instruction Set Computing)	RISC (Reduced Instruction Set Computing)
CPU Vendors	Intel, AMD	AWS Graviton, Apple, Qualcomm
Performance	✔ Strong for general-purpose workloads	✔ Efficient for scale-out and cloud-native workloads
Power Efficiency	✗ Higher power consumption	✔ Lower power usage, ideal for cloud
Cost on AWS	Standard pricing	Often cheaper (e.g., Graviton instances)
Software Compatibility	✔ Broad support (legacy and modern)	● Growing support (some apps may need porting)

Feature	64-bit x86 (x86_64)	64-bit ARM (ARM64 / aarch64)
Best Use Cases	Databases, legacy apps, Windows workloads	Web servers, microservices, containerized apps
Example EC2 Instance	t2.micro, m5.large	t4g.micro, c7g.large
AMI Type	Ubuntu x86_64	Ubuntu ARM64

I AM CURRENTLY USING THE 64 – BIT X86(X86_64) ARCHITECTURE FOR LAUNCHING AN EC2 INSTANCE BECAUSE OF THESE REASONS:-

Reason	Explanation
Wider Compatibility	Works with most AMIs, software, and tutorials — including Ubuntu 22.04
Free Tier Support	t2.micro (x86) is free-tier eligible and widely used
Simpler Setup	No need to worry about ARM-specific packages or compatibility
Better Documentation	Most AWS guides and CLI examples use x86_64
SSH Access	Easier to troubleshoot and configure with standard tools

WE HAVE TO USE 64 – BIT(ARM64/ aarch64) WHEN WE HAVE THESE CONDITIONS LIKE:-

Situation	Why ARM64 Is a Good Fit
Cost-sensitive projects	ARM-based instances (like t4g.micro) are often cheaper than x86 alternatives
Cloud-native apps	Ideal for containers, microservices, and serverless workloads
High-scale deployments	ARM64 offers better performance-per-dollar for horizontal scaling
Modern Linux distros	Ubuntu, Amazon Linux 2, and others support ARM64 out of the box
Learning Graviton architecture	AWS Graviton processors are ARM-based — great for exploring modern cloud hardware
Energy-efficient computing	ARM CPUs consume less power, useful for green computing or long-running workloads

WHEN NOT TO USE ARM64:-

Situation	Why x86_64 Might Be Better
Legacy software	Some older apps or drivers may not support ARM64
Windows workloads	Most Windows AMIs are x86-only
Broad compatibility	x86_64 has wider support across tools, libraries, and tutorials
Quick prototyping	x86_64 is safer when you're unsure about architecture needs

NEW TERMINOLOGY :-

- GRAVITON ARCHITECTURE:-
- **AWS Graviton is a family of 64-bit ARM-based processors designed by AWS to deliver better price-performance and energy efficiency for EC2 instances. It's ideal for modern cloud-native workloads and is available in instance types like t4g, c7g, and r7g.**

What Is AWS Graviton?

- **Architecture:** Based on **ARM64 (aarch64)**, using RISC (Reduced Instruction Set Computing)
- **Designed by:** Annapurna Labs (an AWS subsidiary)
- **Purpose:** Optimized for AWS infrastructure — not sold outside Amazon
- **Generations:**
 - **Graviton1** (2018): Basic workloads (A1 instances)
 - **Graviton2** (2019): 40% better price/performance than x86 (T4g, M6g, C6g)
 - **Graviton3** (2022): 25% faster compute, better for ML and HPC (C7g, R7g)

Benefits of Graviton Architecture

- **Lower cost:** Up to 40% savings over x86-based instances
- **Energy efficient:** Up to 72% less power consumption
- **High performance:** Great for scale-out workloads like containers, web apps, and microservices
- **Free-tier eligible:** t4g.micro available for 750 hours/month until Dec 31, 2025

Limitations

- **Software compatibility:** Some legacy x86 software may not run on ARM64
- **No Windows AMIs:** Graviton is Linux-only
- **Learning curve:** Requires awareness of ARM-specific packages and optimizations

Use Case	Graviton Advantage
Web servers	Lower cost, high throughput
Containers (Docker)	Efficient scaling
CI/CD pipelines	Fast builds, low cost
ML inference	Graviton3 boosts performance

AFTER SELECTING THE AMI IMAGE WHICH IS THE BELOW IMAGE :-

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

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Canonical, Ubuntu, 24.04, amd64 noble image

Architecture	AMI ID	Publish Date	Username	
64-bit (x86)	ami-02d26659fd82cf299	2025-08-21	ubuntu	Verified provider

NEXT WE HAVE TO SLECT THE INSTANCE TYPE:-

- **HERE SINCE MY t2.micro is not eligible under the free plan iam using t3.micro which is in the free tier. WE CAN SEE CLEARLY IN THE BELOW FIGURE**

t2.nano

Family: t2 1 vCPU 0.5 GiB Memory Current generation: true On-Demand Ubuntu Pro base pricing: 0.008 USD per Hour
On-Demand SUSE base pricing: 0.0062 USD per Hour On-Demand Linux base pricing: 0.0062 USD per Hour
On-Demand Windows base pricing: 0.0085 USD per Hour

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.017 USD per Hour
On-Demand RHEL base pricing: 0.0268 USD per Hour On-Demand Linux base pricing: 0.0124 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0142 USD per Hour On-Demand SUSE base pricing: 0.0124 USD per Hour

t2.small

Family: t2 1 vCPU 2 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0248 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour On-Demand SUSE base pricing: 0.0548 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0266 USD per Hour On-Demand Windows base pricing: 0.034 USD per Hour

t2.medium

You launch the instance.

[Create new key pair](#)

This instance type is not eligible under the Free Plan. Upgrade your account plan to access this instance type.

Subnet | [Info](#)

SO NOW IMA USING MY INSTANCE TYPE AS “t3.micro”

DIFFERENCE BETWEEN t3.micro and t2.micro:-

Feature	t2.micro	t3.micro
vCPUs	1	2
Memory	1 GiB	1 GiB
Baseline Performance	Lower	Higher
Burstable CPU	Yes (with CPU credits)	Yes (with CPU credits)
Unlimited Mode	Optional	Optional
Network Performance	Low to Moderate	Up to 5 Gbps
ENA Support	No	Yes
Nitro System	No	Yes (better security & performance)
Pricing	Slightly higher	Slightly lower
Free Tier	Included (in most regions)	Included (where t2.micro is unavailable)

- WHY t3.micro is better ?
- Newer Generation: T3 uses the Nitro system, which improves performance, security, and cost-efficiency.
- More vCPUs: Even though both have 1 GiB RAM, t3.micro has 2 vCPUs, giving it better multitasking capability.
- Lower Cost: T3.micro is often *cheaper* than t2.micro in most regions.
- Better Networking: With ENA and up to 5 Gbps throughput, t3.micro handles traffic-heavy workloads better.

NEW TERMINOLOGY:-

NITRO SYSTEM:-

The Nitro System is a modern virtualization architecture used in newer EC2 instance families like T3, T4g, M5, C5, etc. It replaces older software-based virtualization with dedicated hardware and lightweight hypervisors, offering:

- Faster performance with minimal overhead

- Enhanced security through isolated hardware modules
- Better resource allocation (CPU, memory, networking)
- Support for Elastic Network Adapter (ENA) and NVMe storage

Think of Nitro as AWS's way of saying: "Let's make EC2 instances faster, safer, and more efficient by redesigning how virtualization works."

NVme storage:-

NVMe is a protocol designed specifically for SSDs to maximize speed and efficiency. In AWS, NVMe storage appears in two main forms:

1. EBS Volumes Exposed as NVMe Devices

- On Nitro-based EC2 instances (like t3.micro, m5, c5, etc.), even regular EBS volumes are exposed as NVMe block devices.
- Device names look like /dev/nvme0n1, /dev/nvme1n1, etc.
- Requires the AWS NVMe driver, which is pre-installed in most official AMIs.
- Offers better IOPS and lower latency compared to older Xen-based instances.

2. Local NVMe Instance Store

- Some EC2 types (like i3, c5d, m5d) come with ephemeral NVMe SSDs directly attached to the host.
- These are not persistent—data is lost when the instance stops or terminates.
- Ideal for temporary high-speed storage, like caching or scratch space

ENA(ELASTIC NETWORK ADAPTER):-

- ENA (Elastic Network Adapter) in AWS is a high-performance network interface that enables enhanced networking for EC2 instances, delivering significantly higher bandwidth, lower latency, and lower jitter.

IN AWS WHAT ENA MEANS:-

- ENA (Elastic Network Adapter) is a virtual network interface designed by AWS to support enhanced networking on EC2 instances. It allows instances to achieve:
 - Up to 100 Gbps of network bandwidth
 - Low latency and high packet-per-second (PPS) performance
 - Consistent network performance with minimal jitter
- ENA is supported on Nitro-based EC2 instances, such as t3, t4g, m5, c5, r5, and newer families.

Feature	Benefit
High Bandwidth	Up to 100 Gbps for large-scale workloads
Low Latency	Ideal for real-time apps, gaming, and financial systems
Scalable Performance	Supports millions of packets per second (PPS)
Enhanced Networking	Reduces CPU overhead and improves throughput
SR-IOV Support	Uses Single Root I/O Virtualization for direct access to hardware

WHEN TO USE ENA:-

Use ENA when:

- You need high-throughput networking (e.g., big data, ML training, video streaming).
- You're running performance-sensitive applications.
- You're using Nitro-based EC2 instances.

SO AFTER CHOOSING THE RIGHT INSTANCE TYPE IT LOOKS SOMETHING LIKE THE BELOW IMAGE.

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory

Current generation: true

On-Demand Linux base pricing: 0.0112 USD per Hour

On-Demand SUSE base pricing: 0.0112 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour

Free tier eligible

▼

☒ All generations
 [Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

NOW AFTER SELECTING THE AMI IMAGE AND EC2 INSTANCES WE HAVE TO SLECT THE KEY VALUE PAIR WHICH PROVIDES US THE AUTHENTICATION TO LOG IN VIA SSH TO OUR EC2 INSTACE VIA THERMINAL SO WE MUST PROVIDE A KEY VALUE PAIR.

SO NOW IAM GOGIN TO CREATE A NEW KEY VALUE PAIR FOR MY INSTANCE BY CLICKING ON “create Key-pair”.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Select

 [Create new key pair](#)

AFTER CLICKING ON create new key pair we get this window shown below.

Create key pair ✕

Key pair name

Key pairs allow you to connect to your instance securely.

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ **RSA**
RSA encrypted private and public key pair

☐ **ED25519**
ED25519 encrypted private and public key pair

Private key file format

☒ **.pem**
For use with OpenSSH

☐ **.ppk**
For use with PuTTY

 When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#) 

Cancel Create key pair

HERE IMA GOING TO ENTER MY KEY PAIR NAME AND GOING TO SELECT RSA AND private key file format is .pem file(for use with OpenSSH).

SO some differences between terminologies :-

Difference between rsa and ED25519 WHEN TO USE WHAT:-

DIFFERENCES:-

Feature	RSA	Ed25519
AWS Support	✅ Fully supported across all EC2 instance types and AMIs	✅ Fully supported on modern AMIs (Ubuntu, Amazon Linux 2)
Key Format	.pem (default for AWS)	.pem (also supported)
Key Size	2048–4096 bits	Fixed at 256 bits
Performance	🟡 Slower key generation and SSH handshake	✅ Faster and lightweight
Security	✅ Strong (4096-bit recommended)	✅ Stronger per bit; modern cryptography
Compatibility	✅ Works with all AWS regions and tools	🟡 Requires OpenSSH 6.5+ (most modern systems qualify)
Best Use Case	Legacy systems, tutorials, broad compatibility	Cloud-native setups, faster provisioning

WHEN TO USE AND WHY :- IT IS BASED ON THE SCENARIOS PROVIDED:-

Scenario	Use RSA	Use Ed25519
Launching EC2 in AWS (any AMI)	✅ Works universally	✅ Works with modern AMIs (Ubuntu, Amazon Linux 2)
Using PuTTY on Windows	✅ Required (after converting .pem to .ppk)	🟡 Requires conversion and newer PuTTY version
Following legacy tutorials or scripts	✅ Matches examples and defaults	❌ May cause errors if scripts expect RSA
Working on older systems (OpenSSH < 6.5)	✅ Fully supported	❌ Not supported
Connecting to GitHub or GitLab	✅ Supported	✅ Preferred (GitHub recommends Ed25519)
Wanting faster SSH performance	❌ Slower key generation and handshake	✅ Faster and lightweight

Scenario	Use RSA	Use Ed25519
Concerned about file size or transfer speed	✗ Larger .pem and .pub files	✓ Smaller and efficient
Security-conscious cloud-native setup	✓ Strong (4096-bit recommended)	✓ Stronger per bit; modern cryptography
Quantum resistance (future-proofing)	✗ Vulnerable	🟡 Better but not fully quantum-proof
Using EC2 with Graviton (ARM64)	✓ Works fine	✓ Works fine — ideal for modern setups

WHEN TO USE .PPK AND .PEM AND THEIR DIFFERENCES:-

DIFFERENCES:-

Feature	.pem (Privacy Enhanced Mail)	.ppk (PuTTY Private Key)
Used By	✓ Linux/macOS terminal, AWS CLI, OpenSSH	✓ PuTTY and WinSCP on Windows
Default Format	✓ AWS EC2 key pairs	✗ Must be converted from .pem
File Extension	.pem	.ppk
Encryption Format	Base64-encoded PKCS#8 or PKCS#1	Proprietary PuTTY format
Conversion Needed	✗ No (for Linux/macOS)	✓ Yes (convert .pem to .ppk using PuTTYgen)
SSH Compatibility	✓ Works with ssh command	✓ Works with PuTTY GUI

WHEN TO USE WHAT ? :-

Scenario	Use .pem	Use .ppk
Connecting to EC2 from Linux/macOS terminal	✓ Native support with ssh	✗ Not supported
Using AWS CLI or shell scripts	✓ Required format	✗ Not compatible

Scenario	Use .pem	Use .ppk
Using PuTTY on Windows	✗ Must convert to .ppk	✓ Required format
Using WinSCP for file transfer on Windows	✗ Must convert to .ppk	✓ Required format
Uploading key to AWS during EC2 launch	✓ Upload .pem public key	✗ Not accepted
Automating SSH access in CI/CD pipelines	✓ Works with OpenSSH tools	✗ Not supported
Sharing key with collaborators using Linux	✓ Universally accepted	✗ May require conversion
Using Visual Studio Code Remote SSH	✓ Works with .pem	✗ Not supported
Using legacy Windows tools (pre-OpenSSH)	✗ Must convert	✓ Required
Storing keys securely in AWS Secrets Manager	✓ Compatible	✗ Not recommended

NOW THE KEY-PAIR THAT I WILL BE USING IS “roshan-myinstance”. THEN IAM GOING TO CLICK ON “create Key-pair” in the below image. A key-par will be generated.

Key pair type -> RSA

PRIVATE key file format -> .pem

Key pair name -> roshan-myinstance

Create key pair



Key pair name

Key pairs allow you to connect to your instance securely.

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type



RSA

RSA encrypted private and public key pair



ED25519

ED25519 encrypted private and public key pair

Private key file format



.pem

For use with OpenSSH



.ppk

For use with PuTTY



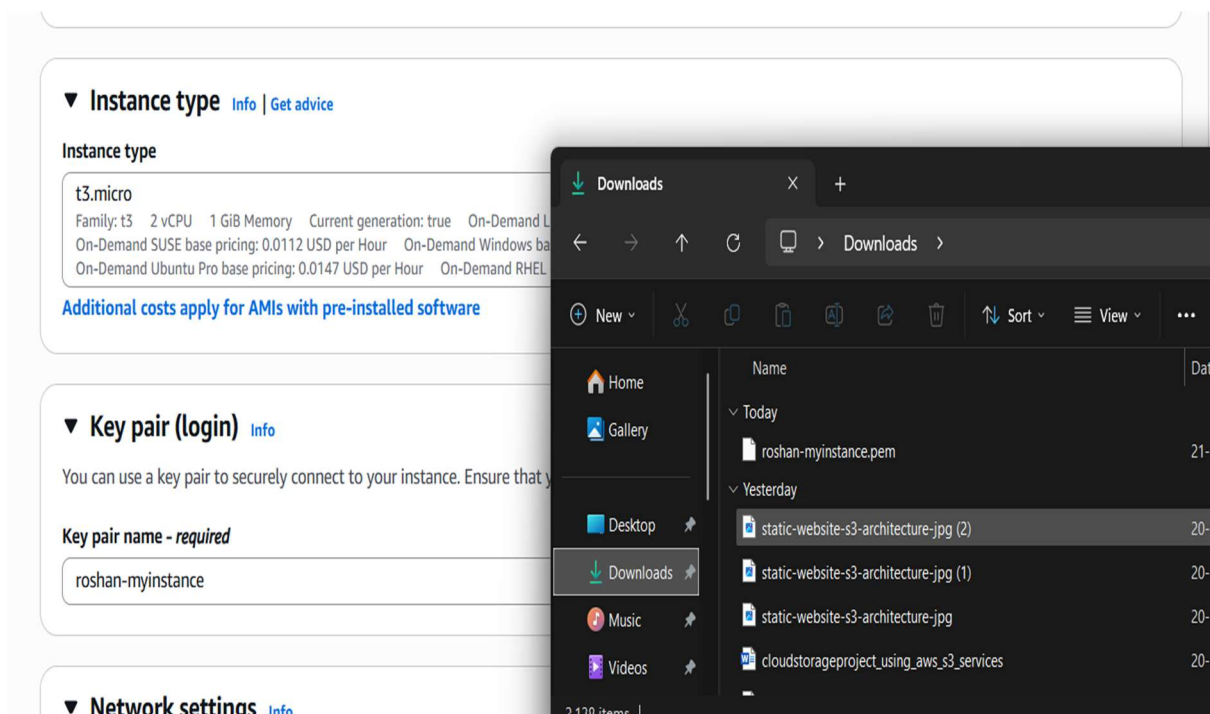
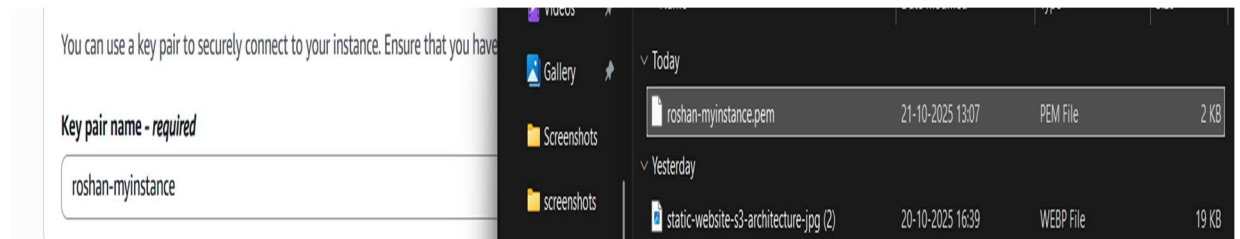
When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#)

Cancel

Create key pair

SELECTING THE RIGHT OPTIONS FOR MY KEY VALUE PAIR IF USING THE .pem FILE THEN GO FOR MOBAXTERM IN WINDOWS IF USING “.PPK” FILE THEN GO FOR PuTTY and u can ssh into the ec2 instance using the key value pair mentioned for the above ec2 instance.

SO NOW WE CAN SEE THAT MY KEY PAIR IS CREATED AND DOWNLOADED IN MY FOLDERS AND AUTOMATICALLY THE KEY PAIR NAME IS FILLED IN AWS MANAGEMENT CONSOLE.



NEXT IS THAT WE WILL SEE THE NETWORK SETTINGS;-

INITIAL NETWORK SETTINGS WITHOUT CHANGING ANYTHING LOOKS LIKE THIS :-

▼ Network settings [Info](#)

[Edit](#)

Network [Info](#)

vpc-062ab221c16cb443d

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

☒ Allow SSH traffic from

Helps you connect to your instance

Anywhere

0.0.0.0/0

☐ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

☐ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

[×](#)

AS PER ELEVATE-LABS IAM TICKING THE BOXES FOR HHTP AND HTTPS TRAFFIC TO FLOW THROUGH MY INSTANCE . THE 0.0.0.0/0 Anywhere means the traffic is taken from multiple places or we can say we are giving access to the ec2 instance to allow all traffic.THIS IS THE CHANGED NETWORK SETTTINGS

AND REST EVERYTHING IAM GOING TO KEEP IT DEFAULT.

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

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- ☒ Allow SSH traffic from
Helps you connect to your instance
- ☒ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server
- ☒ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

Anywhere

0.0.0.0/0

▼

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. ✕

AND AUTO ASSIGN PUBLIC IP SHOULD ALWAYS BE ENABLED SO THAT WE HAVE INTERNET OR THE NETWORK ACCESS.CHECK IF IT IS ENABLED.

Description - *required* [Info](#)

launch-wizard-1 created 2025-10-21T06:27:30.119Z

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type [Info](#)

ssh

▼

Protocol [Info](#)

TCP

Port range [Info](#)

22

Source type [Info](#)

Anywhere

▼

Source [Info](#)

Q Add CIDR, prefix list or security group

0.0.0.0/0 ✕

Description - *optional* [Info](#)

e.g. SSH for admin desktop

Remove

▼ Security group rule 2 (TCP, 443, 0.0.0.0/0)

Type [Info](#)

HTTPS

▼

Protocol [Info](#)

TCP

Port range [Info](#)

443

Source type [Info](#)

Anywhere

▼

Source [Info](#)

Q Add CIDR, prefix list or security group

0.0.0.0/0 ✕

Description - *optional* [Info](#)

e.g. SSH for admin desktop

Remove

▼ Security group rule 3 (TCP, 80, 0.0.0.0/0)

Type [Info](#)

HTTP

▼

Protocol [Info](#)

TCP

Port range [Info](#)

80

Source type [Info](#)

Anywhere

▼

Source [Info](#)

Q Add CIDR, prefix list or security group

0.0.0.0/0 ✕

Description - *optional* [Info](#)

e.g. SSH for admin desktop

Remove

THE ABOVE FIGURE SHOWS THE INBOUND RULES. I HAVE NOT SET UP ANY OUTBOUND TRAFFIC ONLY INBOUND TRAFFIC IS SET UP.

▼ Configure storage Info

Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Add new volume

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

🕒 Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems

Edit

FOR CONFIGURED STORAGE AS SHOWN ABOVE IAM USING GP3 because iam using t3.mciro so if we use t2.micro + gp3 how it is based on performance so these differences are given in the below table :-

Scenario	Use t2.micro + gp3	Use t3.micro + gp3
Free-tier eligibility	✔ Yes (t2.micro is free-tier eligible)	✔ Yes free-tier eligible
Burstable performance	✔ Supports CPU bursting	✔ Supports CPU bursting (more efficient)
Modern workloads (Node.js, Python, Docker)	● Works, but slower	✔ Better performance and networking
Graviton or ARM64 testing	✖ Not supported	● Use t4g.micro instead
Cost-efficiency (paid tier)	● Older generation, less efficient	✔ More cost-effective per vCPU
Network performance	● Lower baseline	✔ Higher baseline + ENA support
Storage performance with gp3	✔ Works fine	✔ Works fine — better with t3's CPU/network
Long-term scalability	✖ Limited	✔ Recommended for future-proofing

▼ Configure storage [Info](#)

Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Add new volume

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

🕒 Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

↻

0 x File systems [Edit](#)

SO IN THE ABOVE FIGURE WE CAN SEE THAT I AM USING 8GiB and using gp3,3000 IOPS(INPUT OUTPUT PER SECOND), NOT ENCRYPTED.

THERE IS ALSO ONE MORE SECTION KNOWN AS ADVANCED DETAILS FOR FURTHER PARAMETERS BUT CURRENTLY FOR JUST LAUNCHING AN EC2 INSTANCE THESE ARE THE MAIN POINTS THAT IS IMPORTANT IN ADVANCED DETAILS WE HAVE THE DIRECTORY THAT IT SHOULD BE DEPLOYED AND ALL OTHER PARAMETERS ARE MENTIONED AS THE BELOW PICTURES GIVEN :-

▼ Advanced details [Info](#)

Domain join directory | [Info](#)

Select

 [Create new directory](#) 

IAM instance profile | [Info](#)

Select

 [Create new IAM profile](#) 

Hostname type | [Info](#)

IP name

DNS Hostname | [Info](#)

- ☒ Enable IP name IPv4 (A record) DNS requests
- ☒ Enable resource-based IPv4 (A record) DNS requests
- ☐ Enable resource-based IPv6 (AAAA record) DNS requests

Instance auto-recovery | [Info](#)

Select

Shutdown behavior | [Info](#)

Stop

Stop - Hibernate behavior | [Info](#)

Select

Termination protection | [Info](#)

Select

Stop protection | [Info](#)

Select

SOME MORE PARAMETERS GIVEN IN THE ABOVE FIGURE FOR AUTO – RECOVERY,HIBERNATE BEHAVIOUR,IF U WANT TO ATTACH ANY IAM INSTANCE PROFILES

Detailed CloudWatch monitoring [Info](#)
Select

Credit specification [Info](#)
Unlimited

Additional charges apply

Placement group [Info](#)
Select [Create new placement group](#)

EBS-optimized instance [Info](#)
Enable

Instance bandwidth configuration [Info](#)
Select

Purchasing option [Info](#)
☒ None
☐ Capacity Blocks
Launch instances for your active capacity blocks
☐ Spot instances
Request Spot Instances at the Spot price, capped at the On-Demand price

Capacity reservation [Info](#)
Select

Tenancy [Info](#)
Select

RAM disk ID [Info](#)
Select

Kernel ID [Info](#)

▼ Summary

Number of instances [Info](#)
1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-02d26659f082cf299

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel [Launch instance](#) [Preview code](#)

SHOWS HOW MUCH CAPACITY U CAN RESERVE PLACEMENT GROUP ETC.

Kernel ID [Info](#)

Select ▼

Nitro Enclave [Info](#)

Select ▼

License configurations [Info](#)

Select ▼

CPU options [Info](#)

Configure CPUs for your instance to optimize performance and save on licensing costs.

☒ Use default CPU options

☐ Specify CPU options

Default active vCPUs

2

Total vCPUs

2

Metadata accessible [Info](#)

Enabled ▼

Metadata IPv6 endpoint [Info](#)

Select ▼

Metadata version [Info](#)

V2 only (token required) ▼

⚠ For V2 requests, you must include a session token in all instance metadata requests.
Applications or agents that use V1 for instance metadata access will break.

Metadata response hop limit [Info](#)

2

Allow tags in metadata [Info](#)

Select ▼

▼ Summary

Number of instances [Info](#)

1

Software image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)

ami-02d26659f882cf299

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

[Preview code](#)

THE ABOVE FIGURE SHOWS U THE VCPUS BEING USED CURRENTLY WE ARE USING 2 VCPUS.

FOR USER DATA SOWIN IN THE BELOW IMAGE

User data - optional | Info

Upload a file with your user data or enter it in the field.

Choose file

☐ User data has already been base64 encoded

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

FINAL SUMMARY OR DETAILS OF MY INSTANCE:-

▼ Summary

Number of instances | [Info](#)

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-02d26659fd82cf299

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

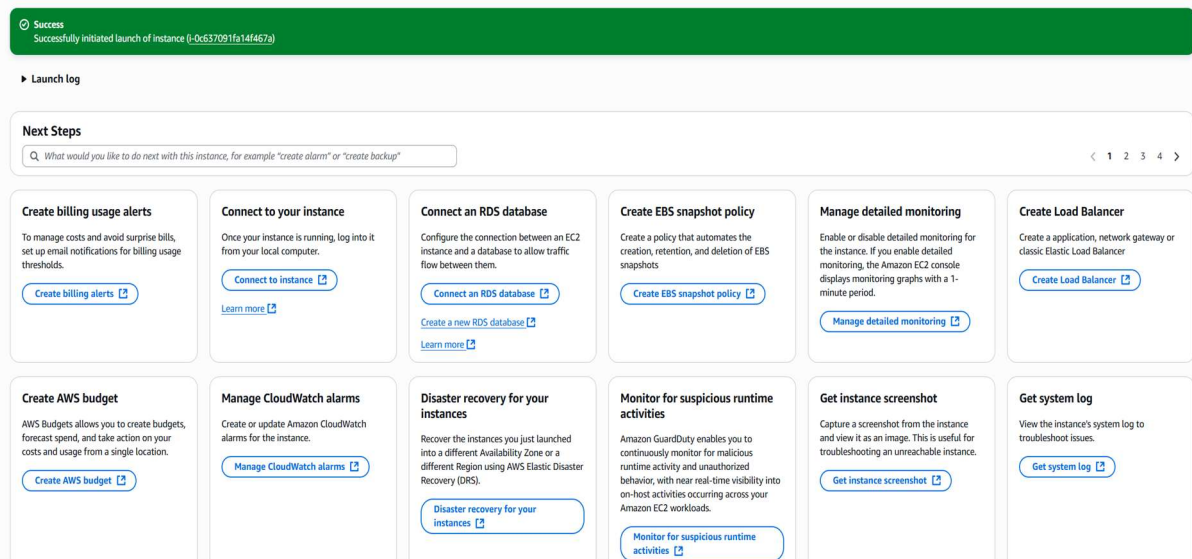
[Cancel](#)

[Launch instance](#)

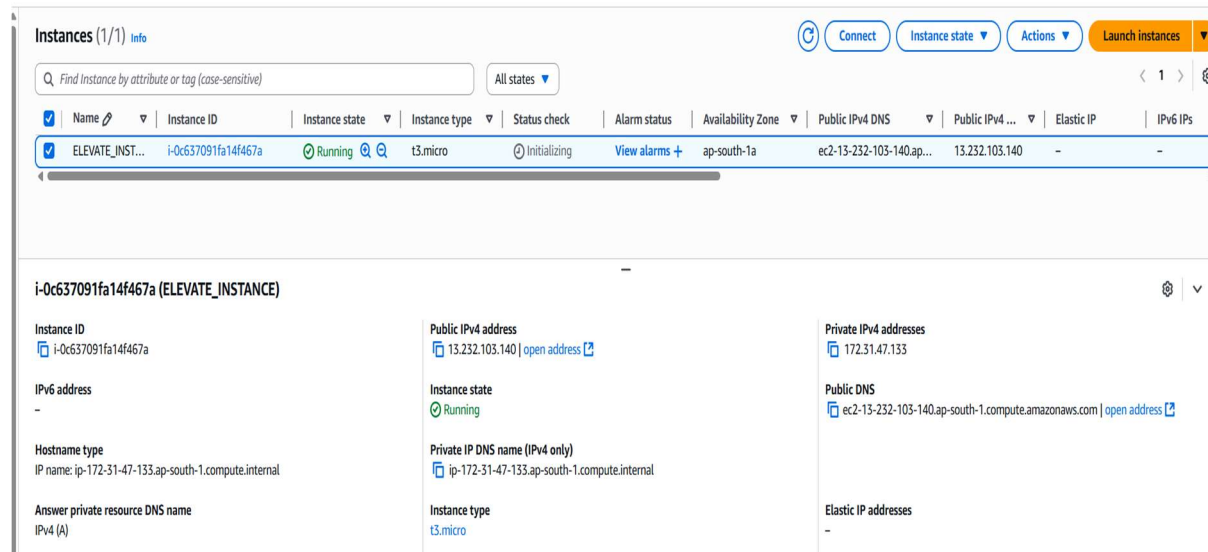
[Preview code](#)

NOW I WILL CLICK ON “Launch instance”.

THIS IMAGE SHOWS I HAVE SUCCESSFULLY LAUNCHED THE EC2 INSTANCE.



WE CAN SEE THAT THE EC2 INSTANCE IS RUNNING:-



NOW IAM GOING TO USE SSH TO GET INTO MY EC2 INSTANCE.I CAN DO THIS IN A NEW BROWSER.

▼ Instance summary [info](#)

Instance ID

[i-0c637091fa14f467a](#)

IPv6 address

-

Public IPv4 address

[13.232.103.140](#) | [open address](#)

Instance state

 Running

Private IPv4 addresses

[172.31.47.133](#)

Public DNS

[ec2-13-232-103-140.ap-south-1.compute.amazonaws.com](#) | [open address](#)

CURRENT PUBLIC IPV4 ADDRESS IS :- 13.232.103.140

CURRENT:-PRIVATE IPV4 ADDRESS:- 172.31.47.133

IAM CLICKING ON CONNECT BUTTON TO CONNECT VIA SSH IN A NEW WEB BROWSER WHEN WE CLICK ON CONNECT THIS WINDOW APPEARS

[EC2](#) > [Instances](#) > [i-0c637091fa14f467a](#) > [Connect to instance](#)

Connect [info](#)

Connect to an instance using the browser-based client.

[EC2 Instance Connect](#) | [Session Manager](#) | [SSH client](#) | [EC2 serial console](#)

Instance ID

[i-0c637091fa14f467a](#) (ELEVATE_INSTANCE)

Connection type

☒ Connect using a Public IP

Connect using a public IP-v4 or IPv6 address

☐ Connect using a Private IP

Connect using a private IP address and a VPC endpoint

☒ Public IPv4 address

[13.232.103.140](#)

☐ IPv6 address

-

Username

Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ubuntu.

[X](#)

Note: In most cases, the default username, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

FOR THE ABOVE WINDOW THAT APPEARED CLICK ON “SSH CLIENT” AND USE THE EXAMPLE URL.

Connect [info](#)

Connect to an instance using the browser-based client.

[EC2 Instance Connect](#) | [Session Manager](#) | [SSH client](#) | [EC2 serial console](#)

Instance ID

[i-0c637091fa14f467a](#) (ELEVATE_INSTANCE)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is roshan-myinstance.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable:
[chmod 400 "roshan-myinstance.pem"](#)
4. Connect to your instance using its Public DNS:
[ec2-13-232-103-140.ap-south-1.compute.amazonaws.com](#)

Example:

[ssh -i "roshan-myinstance.pem" ubuntu@ec2-13-232-103-140.ap-south-1.compute.amazonaws.com](#)

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

BY THE TASK WHAT I GOT TO KNOW IS WE DON'T NEED TO USE ANY SSH CLIENT WE CAN ALSO USE THIS TO LAUNCH AN INSTANCE FROM MOBAXTERM OR PUTTY IF WE HAVE .PEM OR .PPK FILE SINCE I HAVE A .PEM FILE I CAN SHOW IN MOBAXTERM CURRENTLY IAM DOING THE TASK PROVIDED THAT IS

USING THE EC2 INSTANCE CONNECT IAM IN MY REMOTE LINUX SYSTEM.TO DO THIS IMA USING EC2 INSTANCE CONNECT AND CLICKING ON CONNECT BUTTON

Connect

info

Connect to an instance using the browser-based client.

EC2 Instance Connect

Session Manager

SSH client

EC2 serial console

Instance ID

i-0c637091fa14f467a (ELEVATE_INSTANCE)

Connection type

Connect using a Public IP

Connect using a public IPv4 or IPv6 address

Connect using a Private IP

Connect using a private IP address and a VPC endpoint

Public IPv4 address

13.232.103.140

IPV6 address

-

Username

Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ubuntu.

Q ubuntu

X

Note: In most cases, the default username, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel

Connect

WHEN I CLICK ON CONNECT BUTTON A NEWWINDOW OPENS AND IAM INSIDE MY REMOTE LINUX SYSTEM OR IAM IN MY VIRTUAL MACHINE.

ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-0c637091fa14f467a&osUser=ubuntu®ion=ap-south-1&sshPort=22

Search [Alt+S]

Asia Pacific (Mumbai)

Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

* Documentation: <https://help.ubuntu.com>

* Management: <https://landscape.canonical.com>

* Support: <https://ubuntu.com/pro>

System information as of Tue Oct 21 08:27:53 UTC 2025

System load: 0.0 Temperature: -273.1 C

Usage of /: 25.6% of 6.71GB Processes: 109

Memory usage: 22% Users logged in: 0

Swap usage: 0% IPv4 address for ens5: 172.31.47.133

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.

See <https://ubuntu.com/esm> or run: sudo pro status

The list of available updates is more than a week old.

To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;

the exact distribution terms for each program are described in the

individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by

applicable law.

To run a command as administrator (user "root"), use "sudo <command>".

See "man sudo_root" for details.

ubuntu@ip-172-31-47-133:~\$

i-0c637091fa14f467a (ELEVATE_INSTANCE)

PublicIPs: 13.232.103.140 PrivateIPs: 172.31.47.133

NOW IAM INSIDE MY VIRTUAL MACHINE THE OUTPUT LOOKS LIKE THIS:-

```
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Tue Oct 21 08:27:53 UTC 2025

System load:  0.0           Temperature:      -273.1 C
Usage of /:   25.6% of 6.71GB Processes:        109
Memory usage: 22%          Users logged in:  0
Swap usage:   0%           IPv4 address for ens5: 172.31.47.133

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-47-133:~$ ^C
ubuntu@ip-172-31-47-133:~$
```

i-0c637091fa14f467a (ELEVATE_INSTANCE)

PublicIPs: 13.232.103.140 PrivateIPs: 172.31.47.133

THIS IS HOW IT LOOKS THE ABOVE FIGURES SHOW WHILE LAUNCHING THE EC2 INSTANCE IN A NEW BROWSER USING “EC2 INSTANCE CONNECT”

Bash commands:-

uname -a:- gives the system information.

Output is :-

```
ubuntu@ip-172-31-47-133:~$ uname -a
Linux ip-172-31-47-133 6.14.0-1011-aws #11~24.04.1-Ubuntu SMP Fri Aug 1 02:07:25 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
```

SINCE NO FILES ARE THERE IN MY VIRTUAL MACHINE IT WILL NOT SHOW ANYTHING. BUT COMMONLY USED TO LIST FILES , DIRECTORIES ETC.this is the below image

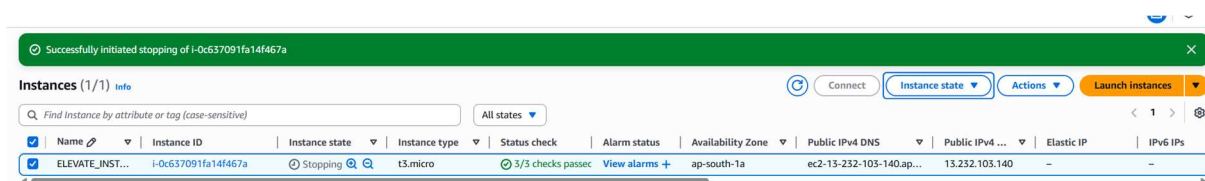
```
ubuntu@ip-172-31-47-133:~$ ls
ubuntu@ip-172-31-47-133:~$
```

Whoami → indicates which system we are currently working in that is ubuntu.

```
ubuntu@ip-172-31-47-133:~$ whoami
ubuntu
```

AND ONE MORE POINT TO NOTICE AS SOON AS WE STOP THE INSTANCE WE CAN NOTICE SOMETHING THAT IS:-

HERE WE HAVBE STOPPED THE INSTANCE AND CONNECT OPTION AUTOMATICALLY GETS DISABLED



ONCE THE INSTANCE IS STOPPED THE PUBLIC IPV4 ADDRESS DISAPPEARS BUT THE PRIVATE IPV4 ADDRESS IS THERE ONLY.

Instances (1/1) [Info](#) Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
ELEVATE_INST...	i-0c637091fa14f467a	Stopped	t3.micro	-	View alarms +	ap-south-1a	-	-	-	-

i-0c637091fa14f467a (ELEVATE_INSTANCE)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

▼ Instance summary [Info](#)

Instance ID i-0c637091fa14f467a	Public IPv4 address -	Private IPv4 addresses 172.31.47.133
------------------------------------	--------------------------	---

NOW OBSERVE THAT AS SOON AS WE START THE INSTANCE WE GET A NEW PUBLIC IPV4 ADDRESS AND THIS IPV4 ADDRESS KEEPS ON CHANGING BUT THE PRIVATE IPV4 ADDRESS REMAINS THE SAME.

NOW ONCE I STARTED RUNNING MY INSTANCE IT WAS IN THE PENDING STATE AND TOOK SOME TIME THEN IT STARTED RUNNING . ONCE WHEN IT STARTED RUNNING WE CAN CLEARLY OBSERVE

STARTING: STATE-

Successfully initiated starting of i-0c637091fa14f467a

Notifications 0 0 2 0 0 0

Instances (1/1) [Info](#) Last updated 2 minutes ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
ELEVATE_INST...	i-0c637091fa14f467a	Pending	t3.micro	-	View alarms +	ap-south-1a	-	-	-	-

i-0c637091fa14f467a (ELEVATE_INSTANCE)

Instance ID i-0c637091fa14f467a	Public IPv4 address -	Private IPv4 addresses 172.31.47.133
IPv6 address -	Instance state Pending	Public DNS -

PENDING STATE;-

Successfully initiated starting of i-0c637091fa14f467a

Notifications 0 0 2 0 0 0

Instances (1/1) [Info](#) Last updated 2 minutes ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
ELEVATE_INST...	i-0c637091fa14f467a	Pending	t3.micro	-	View alarms +	ap-south-1a	-	-	-	-

i-0c637091fa14f467a (ELEVATE_INSTANCE)

Instance ID i-0c637091fa14f467a	Public IPv4 address -	Private IPv4 addresses 172.31.47.133
IPv6 address -	Instance state Pending	Public DNS -

RUNNING STATE

Instances (1/1) [Info](#) Last updated 4 minutes ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

[All states](#)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
ELEVATE_I...	i-0c637091fa14f467a	Running	t3.micro	Initializing	View alarms	ap-south-1a	ec2-35-154-255-175.ap...	35.154.255.175	-	-

i-0c637091fa14f467a (ELEVATE_INSTANCE)

Instance ID
i-0c637091fa14f467a

Public IPv4 address
35.154.255.175 [open address](#)

Private IPv4 addresses
172.31.47.133

Instance state
Running

Public DNS
ec2-35-154-255-175.ap-south-1.compute.amazonaws.com [open address](#)

IPv6 address
-

WE GOT A NEW PUBLIC IPV4 ADDRESS:- 35.154.255.175
THE NEW PRIVATE IPV4 ADDRESS IS SAME AS BEFORE STOPPING THE INSTANCE:- 172.31.47.133

```
System information as of Tue Oct 21 08:27:53 UTC 2025

System load:  0.0           Temperature:    -273.1 C
Usage of /:   25.6% of 6.71GB Processes:      109
Memory usage: 22%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 172.31.47.133

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

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applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-47-133:~$ ^C
ubuntu@ip-172-31-47-133:~$ uname -a
Linux ip-172-31-47-133 6.14.0-1011-aws #11~24.04.1-Ubuntu SMP Fri Aug 1 02:07:25 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
ubuntu@ip-172-31-47-133:~$ ls
ubuntu@ip-172-31-47-133:~$ WHOAMI
WHOAMI: command not found
ubuntu@ip-172-31-47-133:~$ whoami
ubuntu
ubuntu@ip-172-31-47-133:~$ ^C
ubuntu@ip-172-31-47-133:~$
Broadcast message from root@ip-172-31-47-133 (Tue 2025-10-21 08:39:27 UTC):

The system will power off now!
```

i-0c637091fa14f467a (ELEVATE_INSTANCE)

PublicIPs: 13.232.103.140 PrivateIPs: 172.31.47.133

AS SOON AS WE SEE THAT THE SYSTEM IS POWERED OFF WE CANNOT ACCESS OR TYPE ANYTHING WHICH MENS OUR PUBLIC IPV4 ADDRESS HAS CHANGED WIT HTHE NEW ONE WE WILL BE ABLE TO ACCES THE VIRTUAL MACHINE.

```
* Documentation: https://help.ubuntu.com
* Management:   https://landscape.canonical.com
* Support:      https://ubuntu.com/pro

System information as of Tue Oct 21 08:48:16 UTC 2025

System load: 0.02          Temperature:      -273.1 C
Usage of /:  26.1% of 6.71GB Processes:        114
Memory usage: 22%         Users logged in:  0
Swap usage:  0%          IPv4 address for ens5: 172.31.47.133

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Tue Oct 21 08:27:54 2025 from 13.233.177.4
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

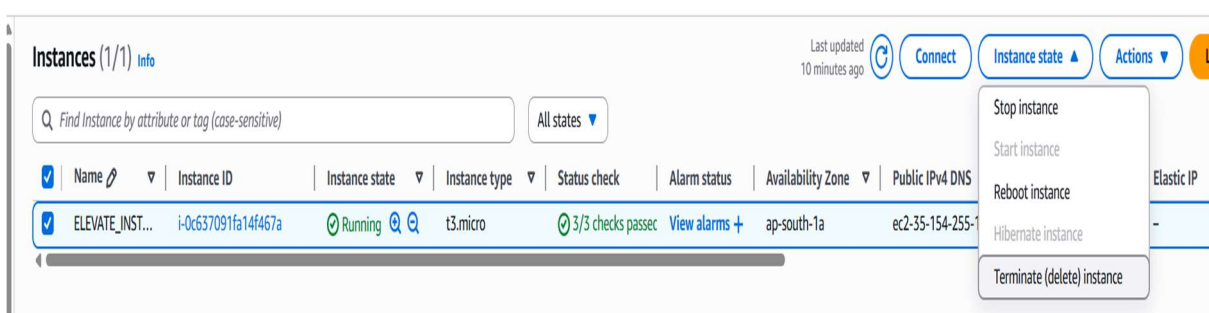
ubuntu@ip-172-31-47-133:~$
```

i-0c637091fa14f467a (ELEVATE_INSTANCE)

PublicIPs: 35.154.255.175 PrivateIPs: 172.31.47.133

HERE IF WE SEE CAREFULLY THE PULIC IPV4 ADDRESS DONE HAS CHANGED AND NOW I WILL BE ABLE TO ACCESS MY VM.

NOW IAM GOING GTO TERMINATE MY INSTANCE:-CLICK ON INSTANCE STATE AND CHOOSE TERMINATE OPTION



BECAUSE FOR EVERY RUN THEIR WILL BE A CERTAIN COST ASSOCIATED TO IT. IT WILL SHOW SHUTTING DOWN INITIALLY AND THEN TERMINATED..

Successfully initiated termination (deletion) of i-0c637091fa14f467a

Instances (1/1) [Info](#)

Last updated 11 minutes ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

< 1 >

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
☒	ELEVATE_INST...	i-0c637091fa14f467a	Shutting-d...	t3.micro	3/3 checks passed	View alarms +	ap-south-1a	ec2-35-154-255-175.ap...	35.154.255.175	-	-

i-0c637091fa14f467a (ELEVATE_INSTANCE)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

BUT DON'T FORGET EVN SOME VOLUMES CAN ALSO BE ATTACHED SO MAKE SURE TO CHECK EVERYTHING BEFORE U LEAVE THE CONSOLE AND DELETE THEM TO UPFRONT OR TO SAVE COSTS IF NOT OFF MUCH USAGE.

SO U CAN SEE THAT MY INSTANCE IS TERMINATED:-

IN STATUS SHOWING TERMINATED.

Instances (1/1) [Info](#)

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

< 1 >

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
☒	ELEVATE_INST...	i-0c637091fa14f467a	Terminated	t3.micro	-	View alarms +	ap-south-1a	-	-	-	-

i-0c637091fa14f467a (ELEVATE_INSTANCE)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

▼ Instance summary [Info](#)

Instance ID
[i-0c637091fa14f467a](#)

IPv6 address
-

Public IPv4 address
-

Instance state
Terminated

Private IPv4 addresses
-

Public DNS
-